

When Structured Evidence Completion Helps—and When It Does Not: A Verified Study of HyperGranular Retrieval-Augmented Generation

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Abstract

Multi-hop retrieval-augmented generation must reconcile evidence completeness with a fixed Top- k context: a relevance ranking can omit a complementary item, whereas expansion can displace evidence that was already useful. We study HyperGranular-RAG, an inference-time label-free evidence-completion framework whose configuration is frozen before each confirmation boundary. It combines adaptive granular balls, query-aware facet hyperedges, and bounded insertion after a protected dense prefix. Across three frozen, closed-candidate HotpotQA and MuSiQue boundaries using Qwen2.5-1.5B-Instruct, the static system repeatedly improved answer F1 over a historical MiniLM dense backbone by +0.01478 [0.00020, 0.02988], +0.01140 [0.00450, 0.01835], and +0.01357 [0.00491, 0.02233]. Removing facet hyperedges reduced performance within the frozen MiniLM system (+0.01336 [0.00341, 0.02343] for FULL minus NoFacet). The boundary changes under a stronger retriever. The original full system underperformed a pre-specified BGE baseline (−0.03998 [−0.05393, −0.02621]), while both cross-space sidecar integration (−0.00256 [−0.00998, 0.00458]) and BGE-native reconstruction (−0.003046 [−0.006880, 0.000631]) were inconclusive relative to BGE. Protected placement was supported only when the inserted set was held identical; its independent BGE-native contrast was inconclusive. Transfer to one official Gemma mobile-QAT configuration was also inconclusive. Post-decision audits further show that small net increases in Gold evidence need not improve answers. Independent verification, Gold isolation, deterministic reruns, and complete reporting of positive, negative, inconclusive, and undefined outcomes support a bounded conclusion: tested structural completion can recover evidence gaps in a compact historical backbone, but no incremental answer-quality value has been established

over strong BGE. The component contrast is against centroid-only ball expansion; neither granular-ball necessity nor hyperedge superiority over a generic diversity/coverage selector has been established.

1 Introduction

Retrieval-augmented generation (RAG) gives language models access to non-parametric evidence at inference time (Lewis et al., 2020; Guu et al., 2020). Dense passage retrieval provides an effective learned relevance ordering (Karpukhin et al., 2020), but multi-hop questions introduce a joint-coverage requirement: the selected context must contain multiple pieces of evidence, not merely high-scoring passages considered independently. HotpotQA and MuSiQue make this requirement explicit through multi-hop questions and supporting-evidence annotations (Yang et al., 2018; Trivedi et al., 2022).

A fixed Top- k budget creates a specific tension. If individually similar units occupy much of the ranking, a complementary bridge may remain outside the generator context. Expanding the ranking can recover that bridge, but every insertion can displace an item already selected by the base retriever. Retrieved evidence is also not uniformly useful to a generator: position, redundancy, prompt construction, and the generator itself affect whether a retrieved item changes the answer (Liu et al., 2024; Ram et al., 2023). Evidence coverage, ranking placement, and answer utility must therefore be measured separately.

HyperGranular-RAG addresses this constrained completion problem. It organizes sentence units into adaptive granular balls, identifies cross-ball relations through query-aware facet hyperedges, and inserts at most four eligible units after a protected Dense Top-10 prefix while preserving an effective Top-20. Retrieval execution is label-free after development-time configuration is frozen: can-

didate construction and ranking do not use answers, supporting-fact labels, generator outputs, or downstream answer scores. Development transactions may use their designated answer labels for configuration selection; confirmation labels remain isolated until rankings and predictions are frozen. Granular balls provide adaptive local organization, while hyperedges represent relations involving more than pairwise adjacency (Xia et al., 2019; Feng et al., 2019).

This paper asks not only whether the method helps, but where its incremental value ends. We evaluate the static system over frozen HotpotQA and MuSiQue closed-candidate boundaries; isolate component contrasts; test one additional generator configuration; bind a pre-specified BGE large-en-v1.5 strong dense baseline; attach the frozen MiniLM-HyperGranular-RAG system as a BGE sidecar; and reconstruct the structural expansion natively in BGE space. Development choices are kept separate from confirmation, and all formal outcomes are independently verified.

The results support three bounded positive claims. First, static HyperGranular-RAG repeatedly improves answer F1 over the historical MiniLM Dense backbone under the frozen Qwen boundaries. Second, facet-conditioned selection improves over the specific centroid-only NoFacet ablation in the frozen MiniLM system. Third, when the cross-space sidecar candidate set is held identical, protected placement is better than direct unprotected placement. The same evidence establishes a strong-retriever boundary: original FULL is clearly below BGE, and neither the cross-space sidecar nor BGE-native reconstruction establishes an answer-quality increment over BGE. We do not claim state-of-the-art performance, universal retrieval improvement, full-wiki validity, broad cross-generator robustness, or an independently isolated granular-ball effect.

Our contributions are:

- a fully specified, inference-time label-free evidence-completion construction under a fixed Top- k budget;
- a multi-boundary evaluation that retains compact-backbone gains and strong-retriever limits in one claim system;
- component evidence that distinguishes candidate construction from placement while exposing unidentifiable and untested contrasts;

- a traceable failure audit and verification protocol that separates evidence gain, displacement, answer utility, and confirmation.

2 Related Work

Dense and multi-hop retrieval. Dense passage retrieval learns dual-encoder representations for efficient passage ranking (Karpukhin et al., 2020). Multi-hop variants extend retrieval from independent passages to evidence chains. Multi-Hop Dense Retrieval conditions later retrieval on preceding evidence (Xiong et al., 2021), while BeamDR uses beam search over dense reasoning paths (Zhao et al., 2021). PullNet iteratively expands from retrieved entities over text and knowledge bases (Sun et al., 2019), and IRCOT interleaves retrieval with chain-of-thought reasoning (Trivedi et al., 2023). HyperGranular-RAG is not an iterative question-rewriting method. Within each frozen closed candidate set, it performs a single bounded completion of an existing ranking.

Graph, hypergraph, and hierarchical retrieval. Structure-aware retrieval represents relations that a flat relevance ordering can miss. RAPTOR retrieves from a tree of recursively summarized text (Sarathi et al., 2024). HippoRAG combines a knowledge graph with personalized propagation (Gutiérrez et al., 2024), and GNN-RAG uses graph-neural retrieval over knowledge graphs (Mavromatis and Karypis, 2025). GraphRAG targets query-focused summarization using graph communities (Edge et al., 2024). HyperGranular-RAG differs in its evidence unit, relation, and budget contract: adaptive balls organize local sentence units; facets define query-conditioned cross-ball hyperedges; and insertion preserves a dense prefix instead of replacing the complete ranking. We do not directly compare against these graph systems.

Evidence expansion and generation. Retrieval-enhanced models can expose external evidence, but the generator still consumes a finite context. RECOMP studies selective and compressed augmentation (Xu et al., 2024), while long-context studies show that evidence position affects model use even when the information is present (Liu et al., 2024). Fusion-in-Decoder demonstrates the importance of how multiple passages are consumed (Izcard and Grave, 2021), and in-context RALM shows that retrieved text can condition language models without parameter updates (Ram

et al., 2023). We therefore use paired answer F1 as the primary endpoint and treat retrieval coverage as supporting evidence.

Alternative completion controls. A simpler completion rule could combine relevance with diversity or uncovered query-term coverage without constructing high-order relations. The No-Facet arm in this study is narrower: it preserves granular-ball geometry and selects up to two eligible balls by query-to-centroid score. It is therefore a matched test of the frozen facet-conditioned selector against centroid-only ball expansion, not a comparison with all relevance–diversity or coverage objectives. We retain this distinction in every component claim.

Evaluation and negative evidence. KILT emphasizes provenance-aware evaluation in knowledge-intensive tasks (Petroni et al., 2021). Reproducibility guidance motivates explicit dependencies, artifacts, and reporting boundaries (Pineau et al., 2021). Statistical uncertainty is not resolved by relabelling an interval crossing zero as “no effect” or equivalence (Amrhein et al., 2019). We retain four evidence states: supported, negative, inconclusive, and not fairly defined.

3 HyperGranular-RAG

3.1 Problem Formulation

For a query q , let $U_q = \{u_i\}$ be the closed candidate set of sentence units. A frozen encoder maps the query and each title + “. ” + sentence unit to L2-normalized vectors. The base relevance score is cosine similarity,

$$s(q, u_i) = \hat{e}_q^\top \hat{e}_i. \quad (1)$$

Ties are resolved by ascending `unit_id`. The Dense baseline returns the first $K_q = \min(20, |U_q|)$ unique units. HyperGranular-RAG constructs an additional ordered set of eligible units but must return the same effective K_q . No answer, Gold supporting unit, or generator output enters this construction.

3.2 Adaptive Granular Balls

Each query’s candidate embeddings begin as one root set. For a current ball B , let

$$c_B = \text{norm} \left(\frac{1}{|B|} \sum_{i \in B} \hat{e}_i \right), \quad d_i = 1 - \hat{e}_i^\top c_B, \quad (2)$$

with radius $r_B = \max_{i \in B} d_i$ and compactness $1 - |B|^{-1} \sum_{i \in B} d_i$. In the compact MiniLM variant, B is split iff its depth is below 6, $|B| \geq 4$, and either $|B| > 3$ or $r_B > 0.78$. The first seed is the unit farthest from c_B ; the second has minimum cosine similarity to the first. Each unit is assigned to the seed with higher cosine similarity. A split is retained only when both children contain at least two units. Ties preserve the frozen candidate order. These constants are frozen implementation parameters, not learned quantities.

3.3 Query-Aware Facet Hyperedges

Each ball aggregates content terms from titles and sentences after frozen ASCII-alphanumeric and stop-word rules. Its facet terms are the intersection between ball terms and query terms. The two balls with highest query-to-centroid cosine become seeds. A non-seed ball can form an expansion hyperedge only if it introduces a new query term and passes frozen gates on ball size, ball score or seed proximity, units per new term, and shared-term redundancy.

For a candidate ball B , let n_B be the number of query terms newly covered beyond the seeds, t_B the total number of query terms in B , m the number of query terms, $b_B = \hat{e}_q^\top c_B$, a_B the maximum cosine to a seed centroid, and ρ_B the fraction of B ’s facet terms already covered. The compact frozen score is

$$\begin{aligned} h_B = & 0.45 n_B / \max(m, 1) \\ & + 0.20 t_B / \max(m, 1) \\ & + 0.20 b_B + 0.15(1 - a_B) \\ & - 0.20 \rho_B - 0.05 |B| / 8. \end{aligned} \quad (3)$$

A candidate must introduce at least one new term, contain at most 8 units, satisfy $b_B \geq 0.12$ or $a_B \geq 0.05$, have at most 6 units per new term, obey $\rho_B \leq 0.90$, and have $h_B \geq 0.10$. Candidates are ordered by descending h_B and then `edge_id`; at most two distinct balls are selected. Units inside an accepted ball remain ordered by original MiniLM cosine and `unit_id`; the frozen unit-level facet bonus is zero. Consequently, the hyperedge selects candidate regions rather than learning a new dense ranker.

3.4 Protected Bounded Insertion

The compact variant protects the first ten unique Dense units. HyperGranular-RAG units whose frozen cosine is at least 0.1957079917192459 are

considered in their frozen order. At most four candidates absent from the protected prefix are inserted immediately after that prefix. Remaining Dense units are appended in their original order, followed only if necessary by unfilled expansion units, and the result is truncated to K_q .

The exact floor is the transferred Stage2E calibration 25th percentile; it is frozen for reproducibility, not presented as an independent optimum (Table 4).

Protection and efficacy are separate claims. Protection limits displacement, but it cannot make a non-complementary candidate useful. This distinction motivates the placement and BGE comparisons below.

3.5 Compact, Sidecar, and Native Variants

The *compact variant* uses MiniLM embeddings for the base ranking, ball geometry, facet gates, and unit ordering. The *cross-space sidecar* keeps BGE Top-20 as the primary ranking while using the frozen MiniLM-HyperGranular-RAG structure to propose eligible candidates; no score fusion is introduced. The *BGE-native variant* reconstructs candidate geometry and structural selection in BGE space under one development-selected, then frozen, configuration. These variants test different scientific questions and are not pooled.

With embeddings precomputed, the closed-candidate procedure is $O(H|U_q|d + |U_q|\log|U_q| + |\mathcal{B}_q|\log|\mathcal{B}_q|)$; Appendix A states the component bounds and memory cost.

4 Experimental Protocol

4.1 Datasets and Frozen Boundaries

HotpotQA provides diverse multi-hop questions and supporting facts (Yang et al., 2018); MuSiQue composes single-hop questions to reduce short-cut solutions (Trivedi et al., 2022). All experiments use benchmark-supplied closed candidate sets. They do not query a full Wikipedia index.

The compact program includes a 1,000-query HotpotQA boundary, a 3,000-query MuSiQue boundary, and a separate 1,000 HotpotQA plus 1,500 MuSiQue component boundary. The cross-space sidecar and BGE-native confirmation studies each use their own frozen, zero-overlap 1,000 HotpotQA plus 1,500 MuSiQue boundary. Selection is deterministic and identity audited.

4.2 Retrievers and Generators

The historical compact backbone is the frozen MiniLM dense retriever. BGE large-en-v1.5 is the pre-specified strong dense baseline; the broader BGE family is documented by Xiao et al. (2023), while the exact model revision is bound in repository manifests.

The main generator is Qwen2.5-1.5B-Instruct FP16 with a fixed short-answer RAG prompt and a 4,096-token input cap. One additional transfer evaluation uses google/gemma-4-E2B-it-qat-mobile-transformers in its official mobile-QAT format. The Qwen-Gemma comparison is deployment-bound: architecture, quantization/numerical format, and runtime effects are not isolated. It cannot support a pure model-architecture ranking or a claim about all devices.

4.3 Development-Confirmation Separation

All method and decision parameters are frozen before the corresponding confirmation boundary is opened. The BGE-native development study selected configuration C10 with a development answer-F1 difference of +0.003143. That value is selection evidence only. It is neither pooled with confirmation nor presented as a confirmed positive result. Automatic BGE-native parameter search stopped after the pre-registered confirmation outcome.

4.4 Metrics and Statistical Decisions

The primary endpoint is paired query-level answer-F1 difference. Answer exact match (EM), retrieval complete recall at 20 (CR@20), and evidence recall at 20 (ER@20) are supporting endpoints. Frozen comparisons use 10,000 paired bootstrap samples and stage-specific pre-specified decision rules. Dataset-joint comparisons assign equal weight to HotpotQA and MuSiQue. An interval crossing zero is INCONCLUSIVE; no equivalence margin or formal equivalence test was specified. Sample sizes define available resource and interval precision, not a formal power guarantee.

4.5 Label-Free Retrieval Execution and Gold Isolation

Inference-time candidate construction, retrieval, prompting, and generation do not read labels or score feedback. Designated development labels may select a configuration (including BGE-native C10), whereas confirmation labels remain inaccessible until rankings, prompts, and predictions

HyperGranular-RAG: protected high-order evidence completion under a fixed Top-k budget

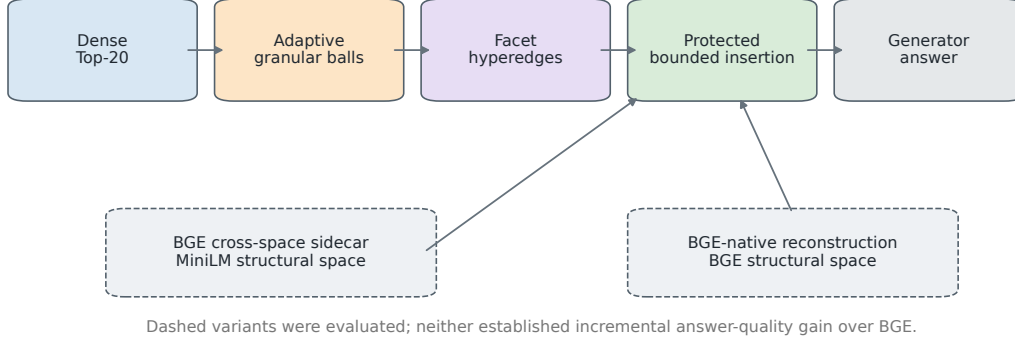


Figure 1: **HyperGranular-RAG method overview.** The compact system organizes locally related evidence as adaptive granular balls, links query-relevant facets across balls with high-order hyperedges, and inserts bounded candidates after a protected prefix under the same Top- k budget. Dashed cross-space and BGE-native variants are evaluated configurations, not validated improvements over BGE.

are frozen and are not reused for selection. Thus “label-free” qualifies retrieval execution after configuration freeze, not the entire research process. Formal evaluators verify Gold identity before parsing it; independent verifiers reconstruct order, metrics, bootstrap summaries, decisions, and artifact identities. Main/rerun identity is checked in full or on a pre-hash deterministic subset.

5 Results

5.1 Repeated Gains over Compact Dense Retrieval

Table 1 and Figure 2 show three frozen Qwen comparisons. On 1,000 HotpotQA queries, FULL minus DENSE answer F1 is $+0.01478$ $[0.00020, 0.02988]$. On 3,000 MuSiQue queries, the difference is $+0.01140$ $[0.00450, 0.01835]$. The separate joint component boundary again supports FULL minus DENSE with a dataset-equal-weight difference of $+0.01357$ $[0.00491, 0.02233]$.

The effects are small but directionally repeated. Their scope is the historical MiniLM backbone, Qwen generator, fixed prompt, closed candidates, and Top-20 budget. They do not establish advantage over modern strong retrievers.

5.2 Component Evidence

Within the frozen MiniLM system, FULL minus NoFacet is $+0.01336$ $[0.00341, 0.02343]$, supporting an incremental role for facet conditioning relative to the specific centroid-only ball selector. It does not establish superiority

over generic relevance–diversity or coverage selection. FULL minus NoProtection is $+0.00354$ $[-0.00675, 0.01389]$, so the independent contribution of protection is inconclusive on that boundary. A flat-unit granular-ball contrast is NOT_FAIRLY_DEFINED: ball seeds, split gates, edge budgets, and radii have no unique matched flat equivalents. The absence of this contrast is not evidence for or against a granular-ball effect.

In the cross-space sidecar, the inserted set is identical between protected and unprotected conditions. PROTECTED minus Unprotected is $+0.01122$ $[0.00129, 0.02104]$, which supports the placement rule within that specific sidecar. However, PROTECTED minus NoFacet is -0.00743 $[-0.01616, 0.00132]$, inconclusive. In BGE-native space, PROTECTED minus Unprotected is $+0.003367$ $[-0.001285, 0.008026]$, and PROTECTED minus NoFacet is $+0.000631$ $[-0.004220, 0.005365]$; both are inconclusive. Component effects are therefore system-dependent.

5.3 Strong-Dense Baseline Boundary

The pre-specified BGE baseline outperforms original FULL. The dataset-equal-weight FULL minus BGE difference is -0.03998 $[-0.05393, -0.02621]$. This negative result is qualitatively different from an uncertain contrast: the complete interval lies below zero. It rules out the claim that the original compact-backbone HyperGranular-RAG system is competitive with this strong BGE

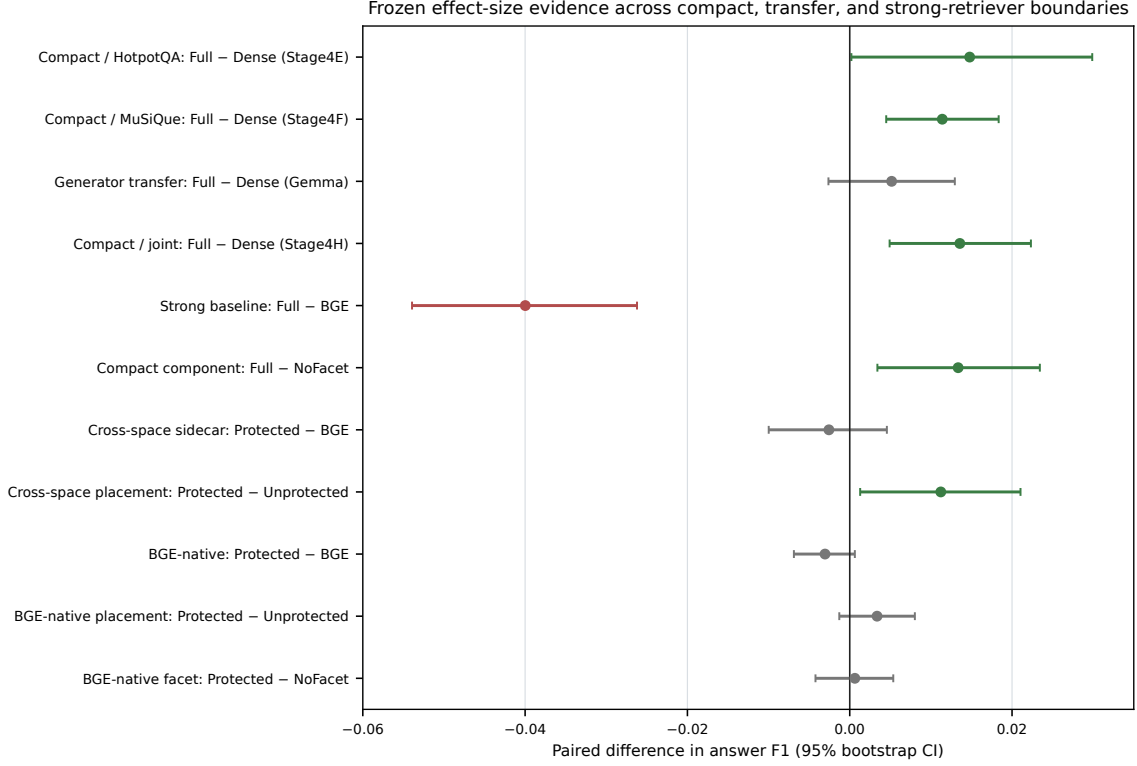


Figure 2: **Frozen answer-F1 effect sizes.** Points are paired answer-F1 differences and bars are pre-specified 95% bootstrap confidence intervals. The zero line is explicit. INCONCLUSIVE means that the interval crosses zero; it is not an equivalence claim.

Boundary	F1 D/F	EM D/F	CR@20 D/F	ER@20 D/F	Avg. ins.	F1 G/H	Δ F1 [95% CI]
HotpotQA	.42150/.43628	.35600/.36600	.73700/.79800	.87860/.90818	1.9300	47/34	+0.01478 [.00020, .02988]
MuSiQue	.13595/.14735	.10033/.11033	.58900/.65000	.80553/.84128	2.4110	125/77	+0.01140 [.00450, .01835]
Joint	.30446/.31803	.25467/.26400	.66617/.72217	.84529/.87657	2.1628	102/73	+0.01357 [.00491, .02233]

Table 1: Absolute compact-dense outcomes and paired effects. D/F denotes DENSE/FULL; G/H counts queries with positive/negative answer-F1 change. Average insertions are structural candidates placed per query. The Joint row is dataset-equal-weighted for absolute metrics; G/H and insertions use all 2,500 queries. All G/H and insertion counts are descriptive reconstructions from frozen rankings and query audits.

Contrast	Δ F1 [95% CI]	State
MiniLM facet	+0.01336 [0.00341, 0.02343]	Supported
MiniLM protection	+0.00354 [-0.00675, 0.01389]	Inconcl.
Sidecar placement	+0.01122 [0.00129, 0.02104]	Supported
Sidecar facet	-0.00743 [-0.01616, 0.00132]	Inconcl.
Native placement	+0.00337 [-0.00129, 0.00803]	Inconcl.
Native facet	+0.00063 [-0.00422, 0.00536]	Inconcl.
Flat-ball control	—	Not defined

Table 2: Complete component evidence. “Native” denotes the BGE-native confirmation boundary; effects are not transported between semantic spaces. The flat-ball row has no matched estimand.

retriever on the frozen boundary.

Two additional baselines on the joint component confirmation boundary also prevent a narrow Dense-only reading. FULL minus BM25 is $-0.00755 [-0.02259, 0.00773]$, and FULL minus

a Dense-BM25 hybrid is $-0.00320 [-0.01591, 0.00942]$. Both are inconclusive and neither supports superiority. BGE remains the decisive strong baseline because the entire FULL-minus-BGE interval is negative.

5.4 Cross-Space Sidecar Evaluation

The cross-space sidecar asks whether the frozen MiniLM structural expansion can add complementary candidates while BGE retains the primary ranking. Its core PROTECTED minus BGE difference is $-0.00256 [-0.00998, 0.00458]$. Because the interval crosses zero, the result does not establish gain, harm, or equivalence. The positive protected-placement contrast does not change this efficacy result: better placement of a fixed inserted

Boundary	Δ F1 [95% CI]	State
Gemma transfer	+0.00516 [−0.00262, 0.01295]	Inconcl.
Full vs. BM25	−0.00755 [−0.02259, 0.00773]	Inconcl.
Full vs. hybrid	−0.00320 [−0.01591, 0.00942]	Inconcl.
Full vs. BGE	−0.03998 [−0.05393, −0.02621]	Negative
Sidecar vs. BGE	−0.00256 [−0.00998, 0.00458]	Inconcl.
Native vs. BGE	−0.00305 [−0.00688, 0.00063]	Inconcl.

Table 3: Strong-retriever boundary and generator transfer. Each row uses its own frozen evaluation boundary; rows are not pooled.

set is not proof that the set improves BGE.

5.5 BGE-Native Reconstruction

The BGE-native study removes the cross-space mismatch by rebuilding candidate geometry and facets in BGE space. Confirmation remains inconclusive. The dataset-equal-weight PROTECTED minus BGE answer-F1 difference is -0.003046 [$-0.006880, 0.000631$]; the EM difference is -0.003667 [$-0.007667, 0.000004$]. Dataset-specific F1 differences are -0.002907 [$-0.008682, 0.002652$] for HotpotQA and -0.003184 [$-0.008486, 0.001863$] for MuSiQue. Both point estimates are slightly negative, but all intervals cross zero. Thus the tested BGE-native reconstruction establishes neither improvement, damage, nor equivalence.

The confirmation result must not be merged with the positive C10 development difference. The development value selected the configuration; the confirmation boundary evaluated it.

5.6 Generator Transfer

Under the single pre-specified Gemma mobile-QAT configuration, the dataset-equal-weight FULL minus DENSE difference is $+0.00516$ [$-0.00262, 0.01295$], inconclusive. This is not evidence that Gemma is unsuitable for RAG, that Qwen has universally superior base capability, or that one architecture is better. The tested model formats differ, and the result is restricted to the recorded RTX 4060 Laptop 8GB, short-answer prompt, 4,096-token cap, and deployable formats.

5.7 Efficiency and Evidence Displacement

Resource values are hardware-bound descriptors rather than cross-device efficiency claims. For the BGE-native confirmation transaction, retrieval reconstruction took 8.544 seconds; BGE and BGE-native PROTECTED generation took 887.650 and 884.198 seconds, respectively; recorded GPU peak memory was 3.592 GiB.

Post-decision Gold audits clarify why coverage and answer quality diverge (Figure 4). In the BGE-native PROTECTED condition, HotpotQA adds 3 Gold units, displaces 2, and has net $+1$; MuSiQue adds 16, displaces 15, and also has net $+1$. Nevertheless, answer-F1 point estimates relative to BGE are negative. In the cross-space sidecar, HotpotQA has 21 added, 29 displaced, and net -8 , whereas MuSiQue has 59 added, 50 displaced, and net $+9$. These counts are post-decision descriptive and cannot be used to retune the method.

6 Discussion

Why compact retrieval can benefit. The repeated MiniLM gains are consistent with a bounded interpretation: when a compact representation leaves recoverable evidence gaps, local candidate organization plus query-conditioned cross-ball expansion can expose complementary units that a flat ranking omits. The joint component confirmation’s facet contrast supports facet-mediated candidate selection within that implementation. This is a synthesis of the frozen evidence, not a universal theorem about compact encoders.

Why the increment contracts under strong BGE. BGE begins with substantially higher answer quality than the original MiniLM system. The original FULL ranking is clearly worse than BGE, and both tested extensions around BGE yield narrow intervals around small negative point estimates. A stronger representation may already rank much of the recoverable evidence highly, leaving less marginal opportunity for structural expansion. Additional units then compete with already strong evidence. The experiment supports this as a study-specific boundary, not as a claim that structure-aware retrieval can never complement strong retrievers.

Evidence addition is not evidence utility. The BGE-native confirmation is especially informative: net Gold increases slightly, but answer quality does not. A Gold-labelled unit can be redundant with retained context, weakly positioned, difficult for the generator to use, or offset by a different displacement. Conversely, a non-Gold unit can affect the answer. CR/ER, Gold transitions, and answer F1 therefore answer different questions.

Placement matters but cannot substitute for efficacy. The cross-space sidecar placement result isolates ranking position because protected and

unprotected conditions contain the same inserted set. It supports preserving high-ranked evidence under a fixed budget. However, the BGE-native confirmation does not independently confirm the placement increment, and neither placement experiment establishes that PROTECTED exceeds BGE.

Facet effects depend on semantic space. Facet-conditioned selection beats the centroid-only No-Facet arm in the frozen MiniLM system, but its increments are inconclusive in cross-space and BGE-native studies. Query-to-ball geometry, seed identities, facet eligibility, and competition with the base ranking all depend on representation space. The MiniLM effect is therefore neither transported to BGE nor interpreted as superiority over untested diversity/coverage selectors.

Failure occurs at selection and displacement layers. Frozen controller audits locate failure at selection and displacement: resource budgeting retained harm more often than gain, the query-level score reversed direction, and a candidate-level probe remained below its advancement gate (Table 6). These post-Gold development results are explanatory, not confirmation; complementary-candidate detection remains open.

Development is not confirmation. The positive BGE-native development result justified opening one frozen confirmation transaction. It did not license retroactive configuration search after confirmation. Continuing to tune on that boundary would convert confirmation into development. Stopping the search is a research-integrity decision, not a claim that no future structure-aware method can work.

Inconclusive is not equivalent. The sidecar, BGE-native, and generator-transfer intervals cross zero. Without a pre-specified equivalence margin and an equivalence test, these results cannot be called equal. They also do not establish harm.

7 Limitations

Evaluation is closed-candidate rather than full-wiki or open-domain; it does not measure indexing, approximate-nearest-neighbor recall, or corpus-scale noise. The repeated positive evidence uses one Qwen generator, and the only additional generator is a single Gemma mobile-QAT configuration. Only one strong BGE backbone is tested. The BGE-native configuration family is fi-

nite and development-selected; untested variants remain unknown.

A scientifically matched flat granular-ball control was not defined, so the granular-ball structure lacks an independent ablation claim. Protected insertion has inconsistent independent evidence: cross-space sidecar placement is supported, while the joint-component and BGE-native protection-related contrasts are inconclusive. Facet evidence is representation- and implementation-dependent. No matched relevance-diversity, maximum-coverage, or other simple completion selector was evaluated, so the hyperedge representation has not been shown necessary or superior to such alternatives.

The HotpotQA training split may have appeared in generator pretraining. The project can guarantee only that its own retrieval and evaluation respected the frozen boundaries. Post-Gold mechanism audits are descriptive, not confirmatory causal analyses. Sample sizes reflect available resources and interval precision, not a formal power design. No inconclusive result is evidence of equivalence.

8 Reproducibility and Integrity

The repository SHA-binds datasets, models, configurations, rankings, predictions, audits, decisions, and final verifications. All confirmation artifacts are frozen; independent verifiers reconstruct metrics and decisions from those artifacts. Every plot has a CSV source and a derived-file manifest, and two manuscript builds are byte-identical. A read-only verifier checks frozen inputs, tables, figures, citations, links, and claim-boundary wording. Licenses are recorded separately; caches and third-party weights are excluded.

9 Conclusion

HyperGranular-RAG repeatedly improves a historical compact MiniLM backbone across three frozen boundaries, and facet selection improves over its centroid-only comparator. This does not establish granular-ball necessity or superiority over generic diversity/coverage selection. The original method underperforms strong BGE, while cross-space and BGE-native extensions establish no incremental answer-quality gain. Placement can help a fixed set but cannot substitute for efficacy. Structural completion helps only within the tested boundaries.

Ethical Considerations

This study uses existing benchmark datasets and publicly documented model artifacts. It does not recruit human participants or collect new personal data. The analysis reports negative, inconclusive, and undefined results alongside positive findings to reduce selective-reporting risk. A venue-specific determination of whether a formal institutional ethics statement is required remains subject to human confirmation.

Data and Code Availability

The identity-exposing development repository records code, configurations, frozen derived evaluation artifacts, figure source data, and verification manifests. It is not linked in this anonymous draft. Original benchmark datasets, model weights, and large local caches are not redistributed and must be obtained from their official sources under their respective terms. An anonymous artifact snapshot and final reuse license remain pending human decisions.

Generative AI Disclosure (Draft)

Generative AI tools were used for code assistance, experiment-workflow assistance, documentation assistance, and language-editing assistance. They are not authors and do not bear academic responsibility. Human authors retain responsibility for the scientific design, evidence selection, verification, claims, and submitted text. Final wording will be reviewed against the selected venue’s policy before submission.

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A Deterministic Procedures and Frozen Parameters

Algorithm 1: adaptive granular-ball construction. Initialize a queue with $(U_q, 0)$. For each ball B at depth z , compute c_B and r_B . If the depth, size, and radius/size gates do not permit a split, emit B . Otherwise choose the first seed as the unit farthest from c_B , choose the second as the unit least similar to the first, and assign each unit to its more similar seed. Candidate-order ties go to the earlier seed. Accept the split only if both children have at least two units; otherwise emit B . Continue until the queue is empty. Stable candidate order and `unit_id` resolve remaining ties.

Algorithm 2: query-aware facet-hyperedge selection. Tokenize the query and every ball with the frozen rules, intersect each ball’s term set with the query-term set, and select the two highest query-to-centroid balls as seeds. Compute newly covered terms and the frozen h_B score for every non-seed ball. Reject balls that fail any eligibility gate. Sort survivors by $(-h_B, \text{edge_id})$, retain at most two distinct balls, and order their units by original MiniLM cosine and `unit_id`. No Gold label or answer score enters this procedure.

Algorithm 3: protected bounded insertion. Let $K_q = \min(20, |U_q|)$ and protect the first $\min(10, K_q)$ unique Dense units. Traverse the eligible expansion order, retain units at or above the frozen score floor, skip protected duplicates, and place at most four units immediately after the prefix. Append unused Dense units in their original order, then expansion units only if needed to fill K_q . Deduplicate stably and truncate to K_q .

Complexity. For $n = |U_q|$, $B = |\mathcal{B}_q|$, embedding dimension d , and maximum depth H , Dense scoring/sorting costs $O(nd + n \log n)$; recursive two-seed assignment costs $O(Hnd)$; facet scoring and ordering costs $O(n + Bd + B \log B)$ apart from tokenization; insertion costs $O(n)$. Working memory is $O(nd + n + B)$. Encoder and generator inference are excluded.

Group	Frozen value	Provenance and sensitivity status
Ball split	depth < 6 ; splittable size ≥ 4 ; radius gate 0.78; minimum child size 2	Frozen compact implementation; no confirmatory sensitivity sweep
Ball seeds and ties	farthest from centroid; then least similar to seed 1; stable candidate order	Deterministic rule, not learned
Facet seeds/budget	2 seed balls; at most 2 expansion balls	Frozen compact implementation; no confirmatory budget sweep
Facet gates	new terms ≥ 1 ; ball size ≤ 8 ; $b_B \geq .12$ or $a_B \geq .05$; units/new term ≤ 6 ; redundancy $\leq .90$; $h_B \geq .10$	Frozen compact implementation
Facet score weights	.45, .20, .20, .15, -.20, -.05; unit bonus 0	Frozen implementation weights; no claim of global optimality
Insertion	protected prefix 10; budget 4; effective $K = 20$	Frozen transferred policy
Candidate floor	0.1957079917192459	Stage2E calibration 25th percentile, transferred exactly; not an independently established optimum
Ordering	descending cosine or h_B , then unit_id or edge_id	Fully deterministic tie-breaking

Table 4: Frozen compact-method parameters. “No confirmatory sensitivity sweep” is a limitation, not evidence of robustness. BGE-native development used a separate finite configuration family and is not pooled with this table.

Applicability boundary of the frozen evidence

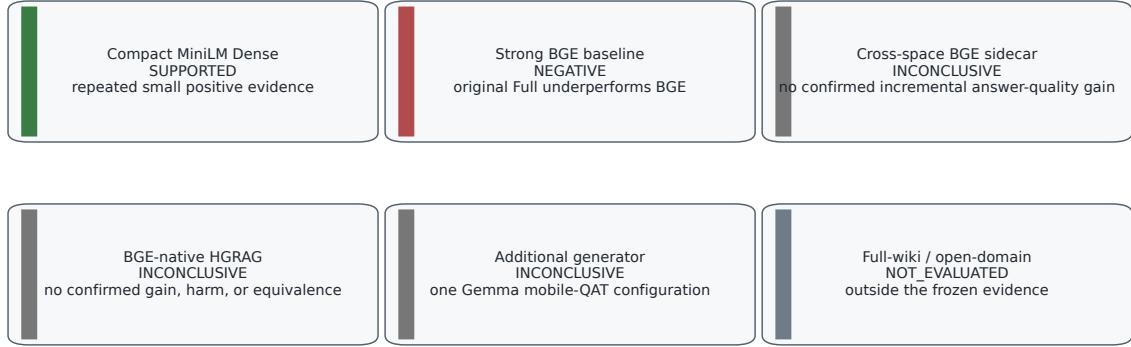


Figure 3: **Applicability boundary.** Repeated small gains are established only for the historical compact MiniLM backbone under closed-candidate Top-20 evaluation. The original system is negative against strong BGE, the tested sidecar and BGE-native extensions are inconclusive, and full-wiki/open-domain operation was not evaluated.

B Applicability Boundary

C Absolute Strong-Retriever Outcomes

D Frozen Failure Audit

E Efficiency and Verification Summary

F Claim–Evidence Boundary Summary

Frozen boundary	Method	F1	EM	CR@20	ER@20
Joint component (Stage4H)	BGE	.35801	.30467	.85400	.93902
Joint component (Stage4H)	Original FULL	.31803	.26400	.72217	.87657
Cross-space sidecar (Stage4I)	BGE	.34301	.28000	.84967	.93594
Cross-space sidecar (Stage4I)	PROTECTED	.34045	.27767	.84350	.93459
BGE-native confirmation (Stage5A)	BGE	.34569	.28500	.86167	.94288
BGE-native confirmation (Stage5A)	PROTECTED	.34264	.28133	.86233	.94288

Table 5: Dataset-equal-weight absolute outcomes on strong-retriever boundaries. Comparisons are valid only within the same frozen boundary; rows from different boundaries are not pooled.

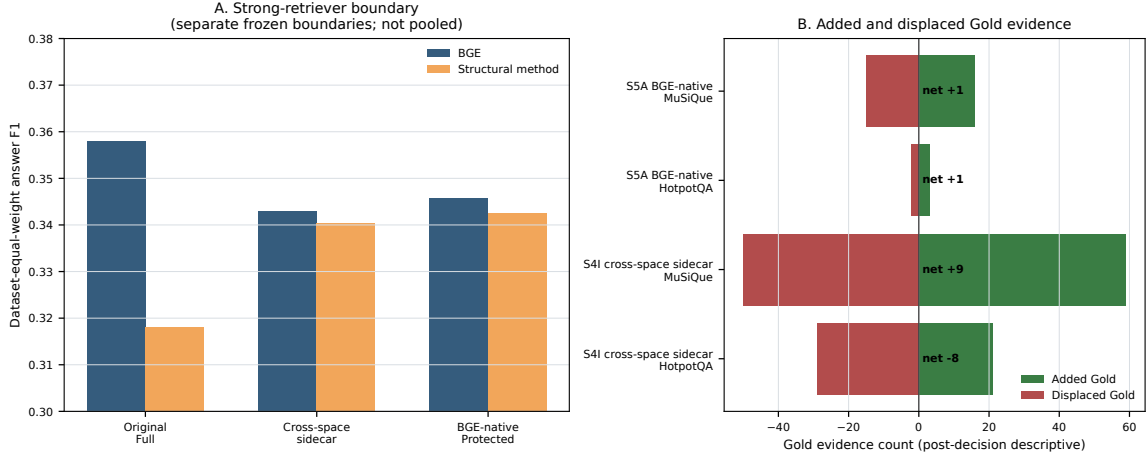


Figure 4: **Strong-retriever boundary and evidence displacement.** Panel A compares each structural method only with BGE inside its own frozen Stage4H, Stage4I, or Stage5A boundary; the groups are not pooled. Panel B reports post-decision descriptive Gold evidence added and displaced. These counts did not control selection or advancement and do not replace end-to-end answer evaluation.

Layer	Frozen observation	Evidential role
Static expansion	Repeated compact-MiniLM answer-F1 gains, but query-level gain and harm coexist	Confirmatory, bounded
Resource controller	Retained 47/94 gain and 53/69 harm queries; retention gap -0.26812 ; two of six gates passed	Valid negative development result
Query score	Gain-vs-harm AUROC 0.39269 [0.30558, 0.48150]	Exploratory, direction reversed
Candidate probe	AUROC 0.64310 [0.55520, 0.72974], AP 0.72198; Brier 0.26321 vs. 0.25338 prevalence baseline	Exploratory, inconclusive
Strong-retriever displacement	Small or mixed net Gold changes did not yield positive answer-F1 relative to BGE	Post-decision descriptive

Table 6: Failure analysis from frozen Stage4B–Stage4D and strong-retriever audits. Exploratory and post-decision rows explain observed failure modes but do not create new confirmatory claims.

Item	Value	Evidential role
Stage5A retrieval reconstruction	8.544 s	Hardware-bound descriptive
Stage5A BGE generation	887.650 s	Hardware-bound descriptive
Stage5A native Protected generation	884.198 s	Hardware-bound descriptive
Stage5A GPU peak memory	3.592 GiB	Hardware-bound descriptive
Bootstrap	10,000 iterations per frozen decision	Confirmatory
Gold policy	Isolated until rankings/predictions were frozen	Integrity
Verification	Independent final verifiers passed	Integrity

Table 7: Efficiency and verification summary. Time and memory values do not support cross-device efficiency claims.

Claim	Evidence state	Boundary
Compact FULL exceeds DENSE on three frozen boundaries	Supported	MiniLM, Qwen, closed-candidate Top-20
Facet increment in the frozen MiniLM system	Supported	Stage4H only
Protected placement with identical sidecar insertions	Supported	Stage4I only
Original FULL exceeds strong BGE	Negative	Stage4H
Cross-space sidecar exceeds BGE	Inconclusive	Stage4I
BGE-native PROTECTED exceeds BGE	Inconclusive	Stage5A
BGE-native placement increment	Inconclusive	Stage5A
BGE-native facet increment	Inconclusive	Stage5A
Additional-generator transfer	Inconclusive	One Gemma mobile-QAT setting
Flat granular-ball control	Not fairly defined	No matched contrast
Granular balls are necessary	Not established	Flat contrast undefined
Facet hyperedges beat generic diversity/coverage	Not tested	No matched alternative selector
Open-domain deployment value	Not tested	Closed candidates only

Table 8: The manuscript’s claim system. Inconclusive rows are not equivalence claims, and system-specific component evidence is not transported across representation spaces.